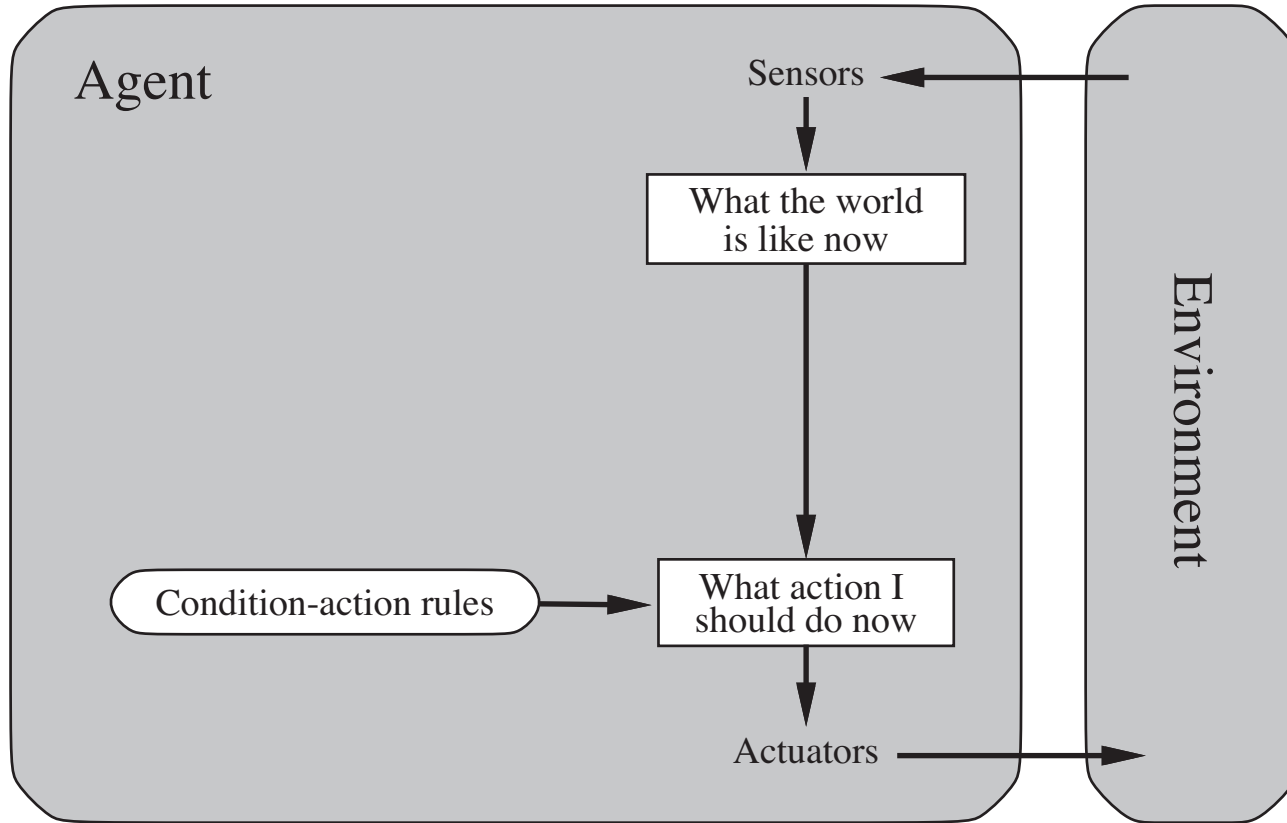


Simple reflex agents



Pacman *agent program* in Python

```
class GoWestAgent(Agent):
```

```
    def getAction(self, percept):
```

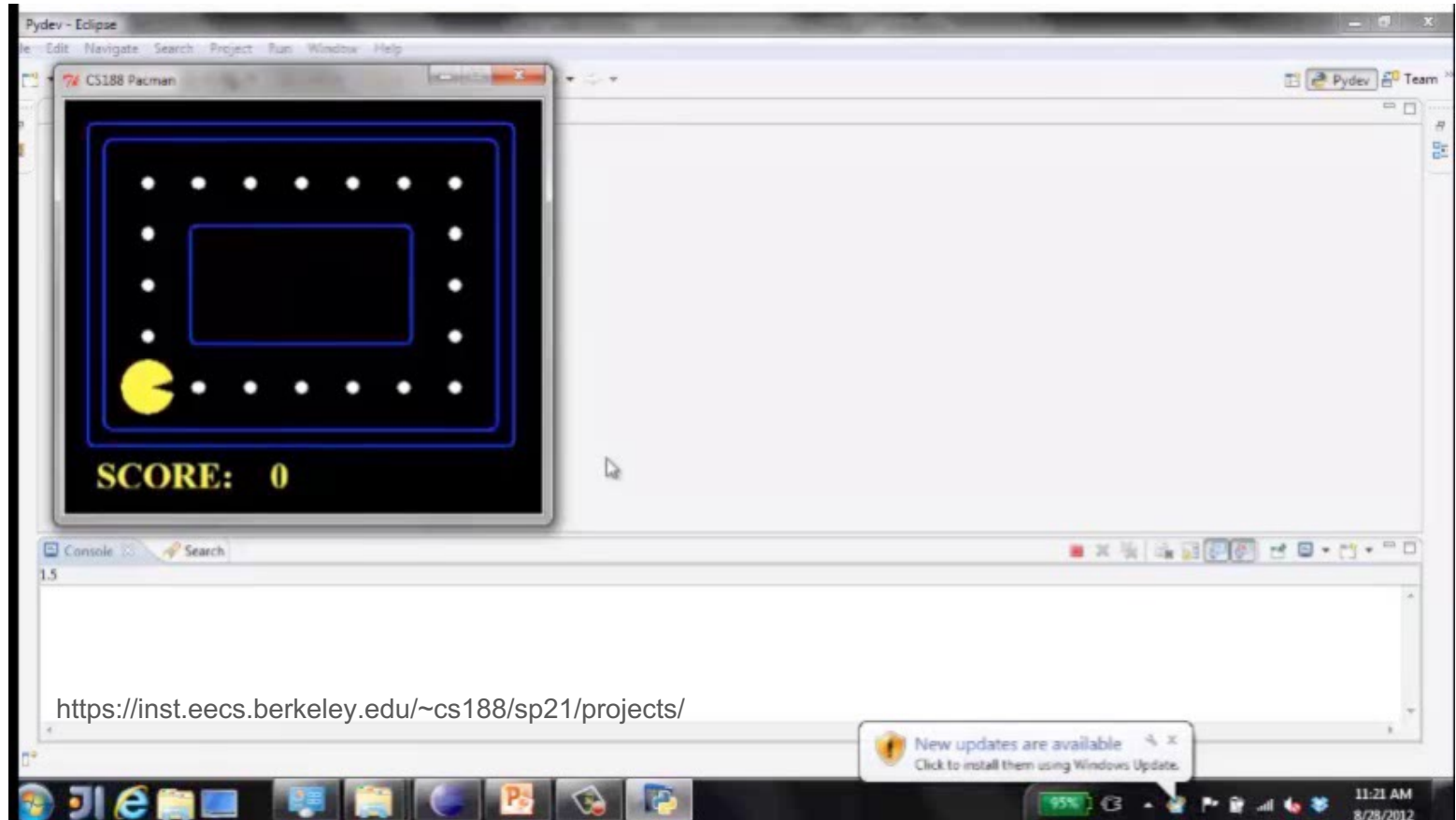
```
        if Directions.WEST in percept.getLegalPacmanActions():
```

```
            return Directions.WEST
```

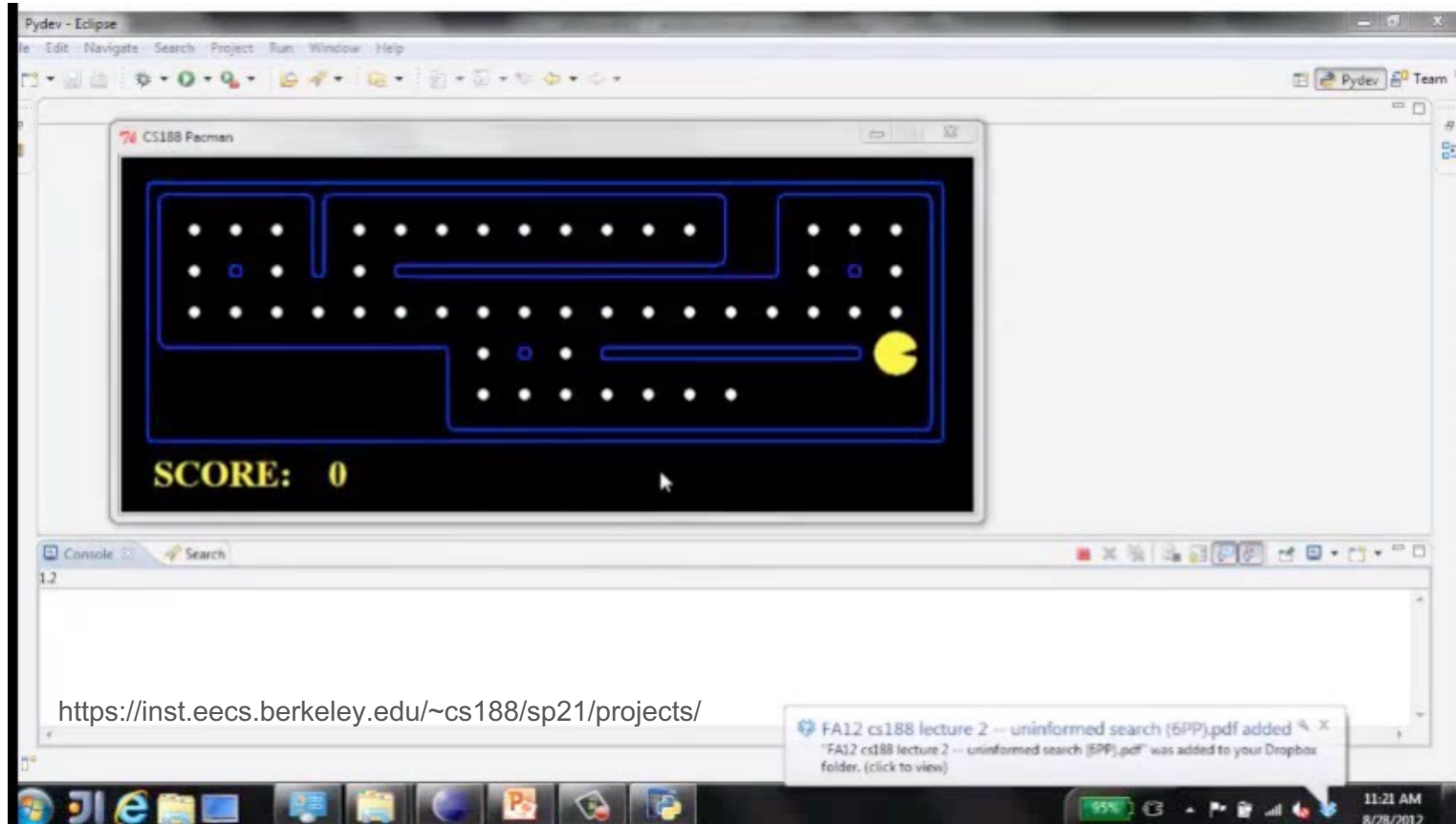
```
        else:
```

```
            return Directions.STOP
```

Eat adjacent dot, if any



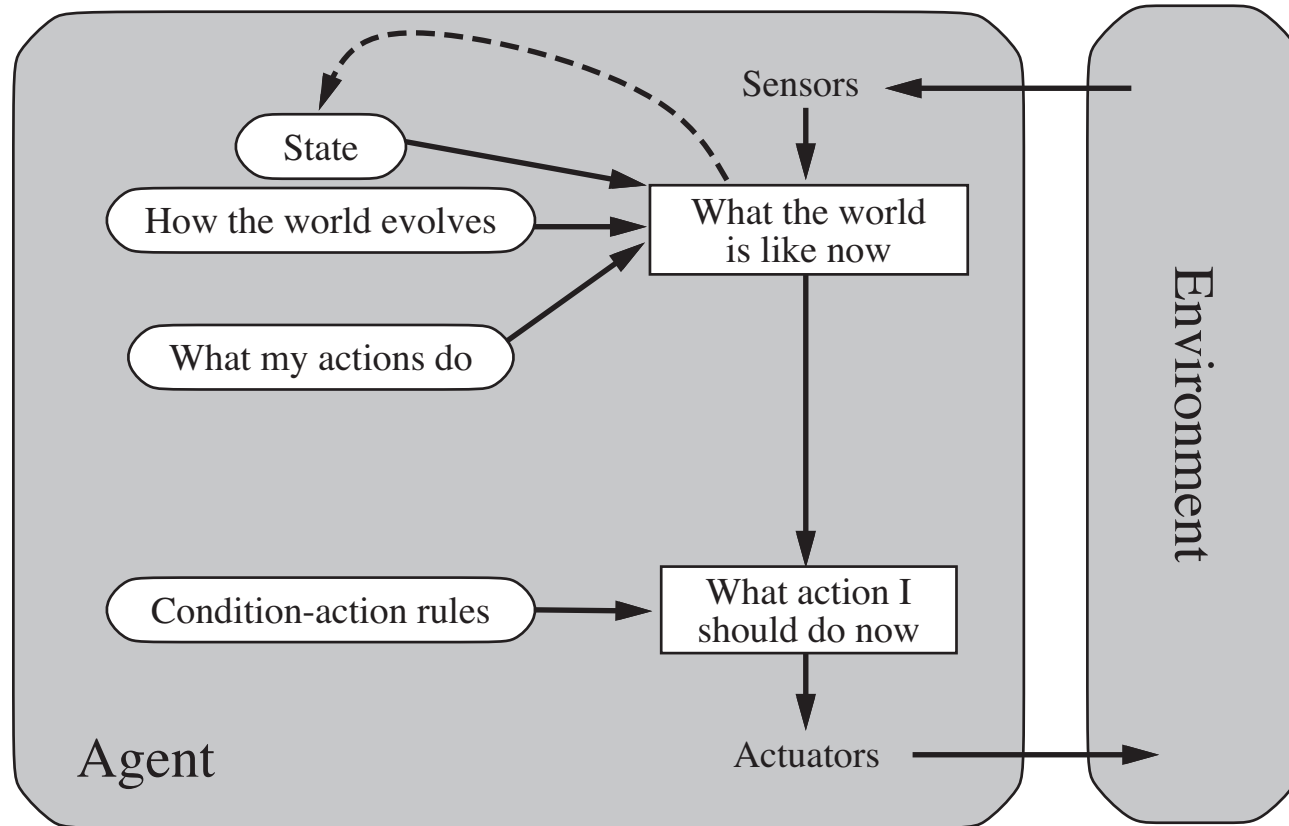
Eat adjacent dot, if any



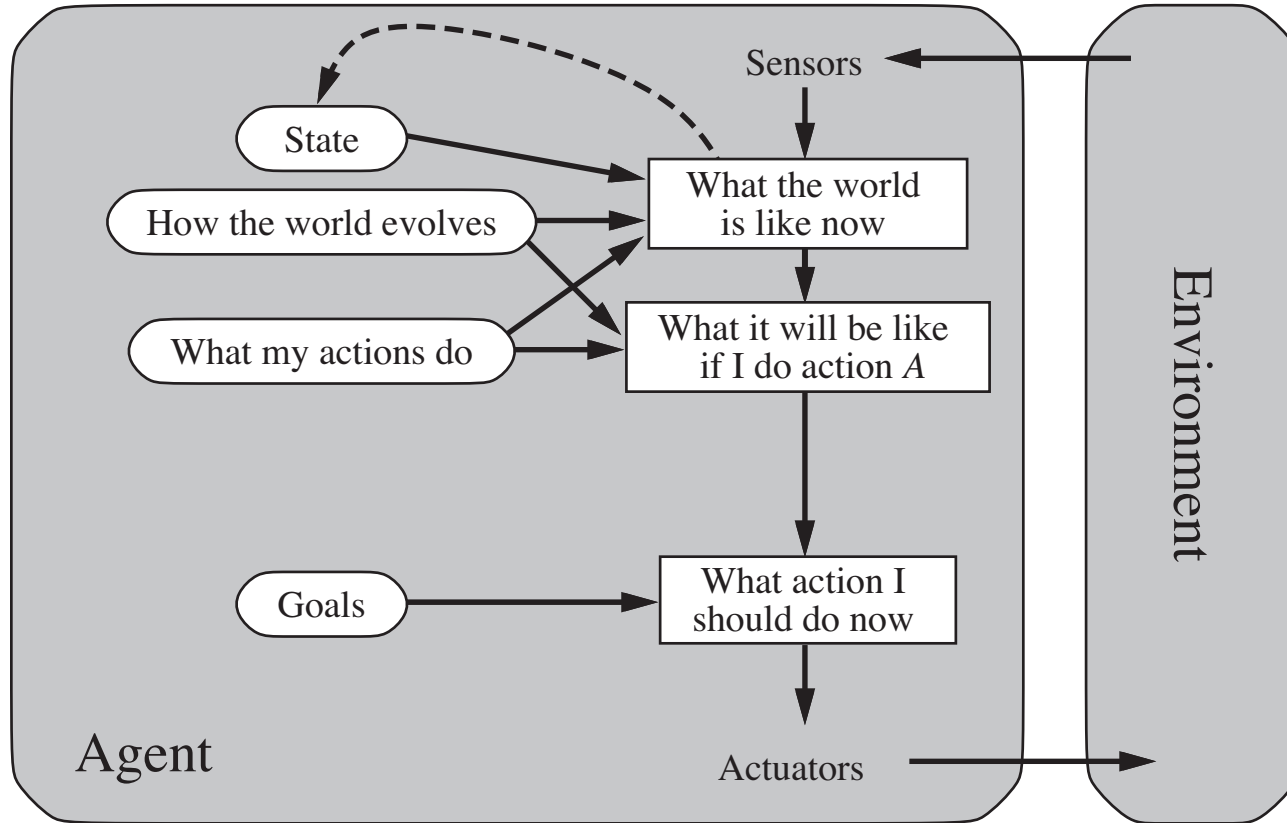
Pacman agent contd.

- Can we (in principle) extend this reflex agent to behave well in all standard Pacman environments?
 - No – Pacman is not quite fully observable (power pellet duration)
 - Otherwise, yes – we can (in principle) make a lookup table.....
 - *How large would it be?*

Reflex agents with state



Goal-based agents



What is AI?

EU vision:

- Artificial intelligence (AI) refers to systems that display intelligent behavior by analyzing their environment and taking actions with some degree of autonomy to achieve specific goals.
- AI-based systems can be purely software-based, acting in the virtual world (e.g., voice assistants, image analysis software, search engines, speech and face recognition systems) or AI can be embedded in hardware devices (e.g., advanced robots, autonomous cars, drones or Internet of Things applications).

Course Description

AI&ML introduces the basics of Artificial Intelligence and Machine Learning.

- This course focuses on transferring knowledge and capability to the "machine" (the intelligent agent) to make it able to understanding the world and answering questions about it, as well as to acting on it to accomplish desired tasks.
- The course will present the basic techniques developed in the field of Artificial Intelligence, concerning modeling the world and exploiting the model for reasoning and planning, focusing on discrete models.
- It will also present basic concepts and techniques from Machine Learning for learning from real world data or from simulations, how to answer and how to act in the world with the purpose of achieving desired goals.

Teaching material:

- [1] Course slides 2022/2023 and additional material.
- [2] Artificial Intelligence: A Modern Approach, Global Edition, 4th Edition by Stuart Russell, Peter Norvig, Pearson 2020 (selected chapters)
- [3] Reinforcement Learning: An Introduction, 2nd Edition by Richard S. Sutton and Andrew G. Barto. MIT Press, Cambridge, MA, 2018. (Freely available online)

Course Contents

Artificial Intelligence (6CFU)

- Introduction to Artificial Intelligence
- Knowledge Representation
- Propositional Logic
 - SAT
 - Tableaux
- First-Order Logic
 - Reasoning in First-Order Logic
 - Tableaux
- Reasoning about Actions
- Deterministic Planning Domains
 - Planning Domain Description Language (PDDL)
 - Planning in Deterministic Planning Domains
 - Search, Heuristics, Best-first, A*
- Nondeterministic Planning Domains (FOND)
 - Planning in Nondeterministic Planning Domains
 - Game Theoretic View
 - Adversarial Search, Search in AND-OR Graphs
- Generalized Forms of Planning
 - Partial Observability
 - Temporally Extended Goals
 - Program Synthesis (Formal Methods)

Machine Learning (3CFU)

- Introduction to Machine Learning
- Supervised Learning
- Deep Learning
- Reinforcement Learning

Propositional Logic

Propositional Logic - Syntax, Semantics and Inferences

Luciano Serafini – Maurizio Lenzerini

FBK, Trento, Italy – Sapienza Università di Roma

Revised and adapted by F. Patrizi

Formal logical system

A **formal logical system** is constituted by several components, including:

- The **language**, with its
 - ▶ syntax (how to form sentences)
 - ▶ semantics (how to assign meaning to sentences)
 - ▶ pragmatics (how to use the logic for a specific objective)

that enables forming the sentences that are acceptable in the formal system

- The **inferencing** apparatus, that enables studying all the relevant relationships between sentences of the language

Propositional logic - Intuition

The first formal logical system that we study is **propositional logic**.

- Propositional logic is the logic of **propositions**
- Intuitively, a proposition is anything that has a truth value (i.e., can be **true** or **false**) in a state of the world (context) of interest.
- The same proposition can be expressed in different ways.

E.g.

- ▶ “B. Obama is drinking a beer”
- ▶ “ The U.S.A. president is drinking a beer”, and
- ▶ “B. Obama si sta facendo una birra”

express the same proposition.

- Warning: obviously, not all sentences in natural language are propositions!
- The language of propositional logic allows us to use propositions and to compose complex sentences (formulas) on the basis of propositions and connectives.

Syntax of propositional logic language: the alphabet

To define the syntax of the propositional logic language, we define the alphabet (set of atomic symbols that can be used in the language), and then the set of sentences that can be formed using the alphabet.

Definition (Propositional alphabet)

Non logical symbols A countable set $\mathcal{P}$ of symbols called **propositional variables**, or **propositional atoms**

Logical symbols (connectives) $\neg, \wedge, \vee, \supset, \rightarrow$ and $\equiv$

Separator symbols "(" and ")"

The intuitive meaning of the connectives are as follows:

- $\neg$ is the negation symbol: $\neg A$ has the opposite truth value as A
- $\wedge$ is the conjunction symbol: $A \wedge B$ is true exactly when both A and B are true
- $\vee$ is the disjunction symbol: $A \vee B$ is true exactly when at least one between A and B is true
- both $\supset$ and $\rightarrow$ denote the material implication symbol: $A \rightarrow B$ means that whenever A is true, B is true as well
- $\equiv$ is the material equivalence symbol: $A \equiv B$ is true if A and B have the same truth value.

The notion of inductive definition

In what follows, we will often define sets and functions **inductively**. An inductive definition of a set S specifies which are the elements of S by using

- one or more **base steps**, that directly specify that some elements belong to S
- one or more **inductive steps**, each of the form **if $x \in S$, then $y \in S$** , where y is related somehow with x .
- a **closure** condition, asserting that nothing else is in S (i.e., anything not deriving from the base or inductive steps is not in S)

Whenever we indicate that a certain definition is inductive, its closure condition is implicit, and therefore we can avoid specifying such condition explicitly.